

Prequential Benchmark of Limit Order Book Mid-Price Movement Prediction Models

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Abstract

Predicting the direction of mid-price movements in high-frequency Limit Order Book (LOB) tick streams, where each tick is an individual trade transaction event, is the most demanding predictive modeling problem in financial machine learning. The tick stream is characterized by near-zero autocorrelation (ACF = 0.046 at lag 1), sample kurtosis of 1,282 (a measure of extreme fat-tailed return distributions) at the one-tick horizon, and microsecond latency requirements that render standard offline learning approaches inapplicable. This paper presents the most comprehensive prequential (test-then-train, zero-lookahead) benchmark of high-frequency LOB mid-price prediction to date: 10 baseline models across 10 trade horizons ($h \in \{1 \dots 500\}$ ticks) evaluated on 11,918,929 BTC perpetual futures (continuously-rolling derivative contracts) ticks under a zero-leakage protocol, with full commercial return, risk-adjusted performance, and deployment efficiency analysis. Against this benchmark, Sticker, a biologically-inspired, gradient-free continual learning model operating at $O(1)$ computational depth, achieves 92.92% non-zero directional accuracy at $h = 1$, a 14 μ s per-tick inference latency, and the maximum commercial signal edge of 0.10 bps (basis points; 1 bps = 0.01% of price) with an annualized Sharpe ratio of 3,913. Sticker establishes a new Pareto frontier (best accuracy at lowest latency simultaneously) across all evaluated architectures, confirming deployment viability from sub-millisecond scalping through sub-second short-swing trading.

Keywords: high-frequency trading, limit order book, prequential evaluation, mid-price movement, continual learning, pareto frontier, scalping, market microstructure

1 Introduction

Predicting the short-term direction of asset mid-prices from Limit Order Book (LOB) tick data is among the most difficult supervised learning problems in quantitative finance. The tick stream is stationary neither in distribution nor in variance:

market regimes shift instantaneously, autocorrelation (serial correlation between consecutive returns) is negligible (ACF = 0.046 at lag 1), and the heavy-tailed return distribution (sample kurtosis 1,282 at $h = 1$) means the overwhelming majority of tick-by-tick price movements are pure microstructure noise (transient bid-ask bounce and order flow friction) rather than directional signal [2, 5]. Standard machine learning models pre trained and evaluated on held-out test sets cannot operate in this environment: distribution shift between training and deployment renders static weights immediately stale, and periodic batch retraining is incompatible with microsecond latency requirements.

The correct evaluation framework for this problem is the prequential (predictive-sequential) protocol [4]: at each tick, the model predicts before observing the future, then adapts using only the resolved label. This protocol prevents lookahead leakage and measures the model’s actual deployed performance rather than its in-sample fit. The closest research [8] evaluate deep models on fixed train/test splits rather than prequential streams, omitting continual adaptation and excluding commercial viability analysis entirely.

The primary contribution of this paper is a comprehensive prequential benchmark on 11,918,929 BTC perpetual futures ticks, evaluating 10 models across 10 trade horizons spanning sub-millisecond scalping ($h = 1$, ~ 0.43 ms) through sub-second short-swing prediction ($h = 500$, ~ 216 ms). The benchmark reports statistical accuracy and full commercial performance: signal edge, Sharpe ratio, maximum drawdown, drawdown recovery speed, break-even fee capacity, and win rate. Against this benchmark, Sticker, a biologically-inspired, gradient-free continual learning model, achieves state-of-the-art directional accuracy at state-of-the-art latency, establishing a new Pareto frontier for deployable high-frequency prediction.

Contributions

1. **High-Frequency Prequential Benchmark:** A zero-leakage prequential streaming benchmark on 11,918,929 BTC LOB perpetual futures ticks is presented, spanning

10 baselines across 10 trade horizons ($h \in \{1 \dots 500\}$) under a strict test-then-train protocol.

2. **Commercial Viability and Tradeability Analysis:** Full financial return dynamics are characterized: signal edge, risk-adjusted performance (Sharpe ratio, Sortino ratio), drawdown recovery speed, and break-even fee capacity, establishing Sticker’s viability across scalping to short-swing trading horizons.
3. **Multi-Horizon Microstructure Regime Characterization:** Empirical return distributions across 10 prediction horizons are characterized, documenting the kurtosis decay from 1282.50 at $h = 1$ to 0.97 at $h = 500$ and the label distribution transition from a Flat-dominated trichotomy to a symmetric binary regime.
4. **Pareto Frontier Continual Predictor:** Sticker is demonstrated as the only evaluated model achieving both highest directional accuracy and microsecond inference latency simultaneously, establishing a new accuracy-latency Pareto frontier for real-time LOB prediction.

2 Background

2.1 Market Microstructure and the Limit Order Book

Financial market microstructure investigates the physical mechanics of price formation under discrete order placement [2, 5]. Modern continuous double auctions operate via a Limit Order Book (LOB), which serves as a real-time ledger recording resting buy orders (*bids*) and sell orders (*asks*) across discrete price levels. The fundamental properties of this ledger include the mid-price $M_t = (p_b + p_a)/2$, acting as the consensus fair value, and the bid-ask spread $S_{\text{spread}} = p_a - p_b$, denoting the fundamental market friction.

Within this framework, market dynamics evolve not in continuous time, but through discrete trade ticks. A trade tick is an individual transaction event generated when an incoming aggressor order (a market order consuming resting liquidity) executes against resting LOB liquidity. Each tick systematically records the microsecond timestamp, trade size, execution price, aggressor direction, and the updated LOB depth. Order Flow Imbalance (OFI), which quantifies net supply-demand shifts in bid and ask liquidity across these interactions, has been established as the dominant driver of short-term price direction [3].

Predicting mid-price movements at the tick level is the most

demanding machine learning subproblem in finance. The tick stream is characterized by near-zero autocorrelation ($\text{ACF} = 0.046$ at lag 1) and staggering heavy-tailed behavior, exhibiting kurtosis of 1,282 at $h = 1$ driven by bid-ask bounce and transient microstructure noise. Combined with instantaneous regime shifts and microsecond latency constraints, these properties render standard pre-trained approaches inapplicable.

2.2 Scalping and Short-Horizon Trading

Scalping refers to a trading strategy that profits from repeated small directional price movements across very short trade horizons ($h = 1$ to $h \approx 20$ ticks), relying on high directional accuracy and low latency rather than large position sizes or long holding periods. At the December 2025 BTC trading velocity of approximately 2,311 ticks per second, $h = 1$ corresponds to ~ 0.43 ms; $h = 20$ ticks to ~ 8.7 ms; $h = 500$ ticks to ~ 216 ms, spanning sub-millisecond scalping to sub-second short-swing prediction. One basis point (bps) = 0.01% of price; all financial metrics are reported in basis points throughout.

2.3 Literature: LOB Prediction Models

The LOB mid-price prediction literature divides into classical econometric models and deep learning approaches. Classical approaches include OFI-based linear regression [3], Kalman filter state-space estimators, ARIMA (AutoRegressive Integrated Moving Average) autoregressive predictors, and GARCH (Generalized AutoRegressive Conditional Heteroskedasticity) variance models. While these achieve high throughput, they are constrained to linear feature combinations and cannot exploit multi-scale order flow interactions.

The deep learning benchmark for LOB prediction is DeepLOB [8], a 2D CNN + Inception + LSTM architecture trained on fixed LOB snapshots. Subsequent work includes selective state-space models (Mamba S6) and transformer-based approaches. A fundamental limitation shared by all deep architectures is their offline training assumption: they require periodic batch retraining and suffer catastrophic forgetting under continual distribution shift, making them incompatible with strict prequential deployment.

3 Sticker Overview

Sticker is a biologically-inspired, gradient-free continual learning model introduced in this research, designed for lowest-latency prediction on stochastic non-stationary tick streams.

Its design is abstracted here to the properties most relevant to financial deployment; the architectural specification is available in the full technical paper.

Structurally, Sticker operates as a flat, non-sequential consensus mapping model. Rather than stacking sequential layers or maintaining recurrent state, Sticker streams input features through N parallel experts through biological attention. This flat graph structure achieves $O(1)$ computational depth: forward inference consists of a concurrent pass per expert followed by a consensus output step, with no sequential layer dependency.

Parameter adaptation is governed by a local biological plasticity rule, node-level outer-product updates that require no backpropagation, no automatic differentiation graph, and no fixed training phase. The model is live and continual learning: it updates in place after each resolved prediction tick.

The integrated Slope-Sign Loss evaluates prediction correctness against directional sign and margin rather than numerical magnitude, preventing the representation collapse and mean-reversion behavior that afflicts standard loss objectives under heavy-tailed noise.

The combined effect of these design decisions is a model that:

1. Processes each tick in 14 μ s inference latency on standard CPU only infrastructure.
2. Adapts to distribution shift in 2.5 μ s per tick via gradient-free local updates 19,280 $\times$ faster than DeepLOB’s back-propagation step.
3. Maintains predictive stability across the full 11.9M tick stream without retraining, batch refits, or hyperparameter tuning.

4 Evaluation

Models are evaluated using standard continual prequential test on high-frequency market microstructure order flow dataset.

4.1 Problem Formulation

Let $\{(x_t, y_t)\}_{t \geq 1}$ be a continuous stream of multivariate state vectors and target directional labels generated by a non-stationary continual process. The vector $x_t \in \mathbb{R}^d$ represents a multi-scale present-state snapshot. In ticker streams, x_t encodes a 56-dimensional snapshot spanning multi-scale order flow imbalances, VWAP spreads, volatility proxies, and

momentum indicators, normalized incrementally online via Welford’s algorithm [7] to eliminate temporal lookahead bias.

The target objective is the directional sign of subsequent movement over a lookahead horizon $h \in \mathbb{Z}_{>0}$:

$$\Delta M_{t,h} = M_{t+h} - M_t, \quad (1)$$

$$y_{t+h} = \text{sign}(\Delta M_{t,h}) \in \{-1, 0, +1\}, \quad (2)$$

where M_t denotes the mid-price at tick t .

Definition (Continual Prequential Test). At each continual event t :

1. Feature vector $x_t \in \mathbb{R}^d$ is computed strictly from events up to tick t .
2. The model emits prediction $\hat{y}_t = f_\theta(x_t)$ prior to observing future states.
3. Prediction $\hat{y}_t$ is enqueued into a causal FIFO buffer of depth h .
4. At tick $t + h$, label y_{t+h} resolves, continual metrics update, and the model executes an online parameter update using pair (x_t, y_{t+h}) .

This prequential procedure prevents lookahead leakage by resolving labels strictly after predictions are emitted.

4.2 Dataset

The test dataset comprises the Hyperliquid BTC Perpetual Futures Level-4 Order Flow Dataset (December 2025), comprising $N = 11,918,929$ continuous trade ticks [1]. The dataset captures 4 discrete LOB depth levels ($k \in \{1, 2, 3, 4\}$) across bid and ask queues, recording microsecond execution timestamps, transaction prices, trade sizes, aggressor side indicators, and multi-level bid/ask order book quotes.

As illustrated in Figure 1, the 11.9M tick stream spans 3 distinct market regimes identified on rolling microstructure volatility and order flow imbalance:

1. **Low-Vol Consolidation:** (24.21% duration) Range-bound price consolidation with baseline volatility (4.23 bps) and high regime persistence (88.77% self-transition probability).
2. **Moderate Trend:** (68.02% duration) The dominant continual regime, featuring steady directional price propagation, baseline volatility (4.45 bps), and strong regime persistence (95.99% self-transition probability).

3. **High-Vol Extreme Shock:** (7.77% duration) Violent liquidity dislocations characterized by a +38% volatility surge (5.86 bps) and elevated shock persistence (88.71% self-transition probability).

Furthermore, empirical return distributions exhibit near-zero price return autocorrelation (ACF = 0.046 at lag 1) and extreme heavy-tailed kurtosis at microsecond trade horizons (Table 1).

Horizon	$h = 1$	$h = 2$	$h = 5$	$h = 20$	$h = 500$
Kurtosis	1282.50	585.22	279.73	62.65	0.97

Table 1: Systematic decay of empirical return kurtosis across lookahead trade horizons $h \in \{1, 2, 5, 20, 500\}$.

At $h = 1$ tick, sample kurtosis reaches Kurtosis = 1282.50 (skewness of -1.442 , return standard deviation of 0.168 bps). As the lookahead horizon expands, return kurtosis decays systematically as documented in Table 1, converging toward near-Gaussian behavior (0.97 at $h = 500$). This extreme next-tick kurtosis quantitatively proves the heavy unlearnable noise governing raw tick feeds, demanding directional optimization over magnitude fitting.

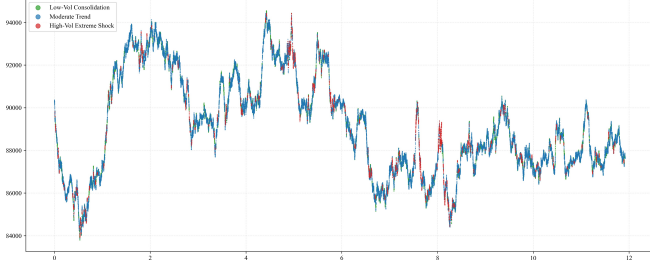


Figure 1: Full-month mid-price trajectory across December 2025 (11.9M continuous trade ticks) partitioned by market regime classification (Low-Vol Consolidation, Moderate Trend, High-Vol Extreme Shock). The y-axis displays BTC mid price in USD, and the x-axis displays trade ticks in millions.

4.3 Feature Engineering

Feature vectors $x_t \in \mathbb{R}^{56}$ are constructed using 56 multi-scale microstructure indicators partitioned across 5 physical feature categories:

1. **Multi-Scale Returns (15 features):** Logarithmic price differences over 5 lookback windows ($\Delta t \in \{1, 5, 10, 20, 50\}$ ticks) across mid, bid, and ask price

streams. *How much has price moved over the last 1, 5, 10, 20, and 50 trades?*

2. **Aggressor Trade Ratios (10 features):** Volume ratios of buyer-initiated trades to total volume over 5 rolling tick windows ($w \in \{1, 5, 10, 20, 50\}$), measuring instantaneous buying versus selling pressure. *Are buyers or sellers more aggressive at each timescale?*
3. **Volatility Proxies (10 features):** Rolling sample standard deviations of tick returns across 5 tick windows ($w \in \{1, 5, 10, 20, 50\}$), quantifying local market noise. *How noisy is the market at short, medium, and longer timescales?*
4. **VWAP Spreads (10 features):** Percentage displacement between instantaneous mid-price and Volume-Weighted Average Price across 5 rolling windows, indicating price displacement from volume-weighted fair value. *Is the current price displaced above or below its recent volume-weighted average?*
5. **Top-of-Book Order Flow Imbalance (11 features):** Net supply-demand order volume differences across top LOB depth levels ($k \in \{1, 2, 3, 4\}$), measuring immediate order book imbalance. *Is more order volume resting on the buy side or sell side of the book at each LOB depth?*

All 56 features are standardized incrementally using Welford’s algorithm [7], preserving strict prequential isolation.

4.4 Multi-Horizon Testing

Models are evaluated across 10 lookahead trade horizons $h \in \{1, 2, 3, 5, 10, 20, 50, 100, 200, 500\}$ ticks. Directional performance is evaluated on non-zero movement ticks ($\Delta M_{t,h} \neq 0$):

$$DA_{nz} = \frac{\sum_t \mathbb{I}(\text{sign}(\hat{y}_t) == y_{t+h} \wedge y_{t+h} \neq 0)}{\sum_t \mathbb{I}(y_{t+h} \neq 0)}. \quad (3)$$

Across multi-horizon testing ($h \in \{1 \dots 500\}$), directional class distributions undergo a smooth structural transition as market-microstructure friction (bid-ask spread bounce) decays. At $h = 1$ tick, zero-movement Flat ticks dominate the stream (65.28%, 7.78M ticks), while Up (17.15%) and Down (17.57%) are perfectly symmetric. As the lookahead horizon extends, Flat tick frequency decreases rapidly: 27.97% at $h = 5$, 9.66% at $h = 20$, 3.18% at $h = 50$, down to 0.36% at $h = 500$ (42,361 ticks), where Up (49.59%) and Down (50.05%) converge to an exact 50/50 binary classification regime.

Evaluation metrics also span statistical classification metrics (Matthews Correlation Coefficient, F1-scores, precision, recall), financial return metrics (average edge per tick in bps, gross/net PnL across fee tiers), risk-adjusted performance (Sharpe ratio, maximum drawdown in bps and duration), trade execution statistics (win rate, profit factor, win-to-loss ratio), and system efficiency metrics (per-tick inference latency quantiles, TPS throughput, parameter counts, memory overhead).

4.5 Baselines

Sticker is evaluated against 9 baseline models spanning classical econometric approaches, linear classifiers, gradient-boosted trees, and deep sequential learning architectures. One basis point (bps) = 0.01% of price; all financial metrics are reported in basis points throughout.

1. **Persistence:** (momentum-only baseline) $\hat{y}_t = \text{sign}(M_t - M_{t-1})$. Follows the naive direction of the most recent price movement.
2. **Linear Regression:** (academic OFI impact model, Cont et al. 2014) 1D Order Flow Imbalance (OFI) impact linear regression model [3].
3. **Kalman Filter:** (optimal linear estimator under Gaussian noise) Kinematic price-velocity state-space estimator.
4. **ARIMA:** (classical autoregressive time-series model) Recursive Least Squares (RLS) AR(5) return predictor.
5. **GARCH:** (conditional variance model; tracks volatility, not direction) Models variance using one lagged squared return shock (ARCH) and one lagged forecast variance (GARCH).
6. **Logistic Regression:** (online linear classifier; strongest non-neural continual baseline) 56-dimensional continual stochastic gradient descent (SGD) logistic regression model.
7. **LightGBM:** (gradient-boosted decision tree; requires periodic batch refit) LightGBM / GBDT based histogram gradient-boosted decision trees.
8. **DeepLOB:** (CNN-LSTM hybrid for limit order book; the deep learning LOB benchmark) 2D CNN + Inception + LSTM [8].
9. **Mamba (S6):** (selective state space model; most recent deep sequence architecture) Selective State Space Model (SSM) S6 architecture [6].

5 Results

5.1 Scalping Accuracy

The directional prediction performance of Sticker is evaluated under the continual prequential test across trade horizons $h \in \{1, 2, 5, 10, 20\}$ trade ticks on the 11,918,880 tick stream. Table 2 presents the statistical classification and accuracy metrics. DA_{nz} (non-zero directional accuracy) measures correctness only on ticks where price actually moves; MCC (Matthews Correlation Coefficient, range -1 to $+1$) is a balanced accuracy measure robust to class imbalance; F1-Macro is the unweighted mean of per-class F1 scores.

Achieving 92.92% directional accuracy on a stream with sample kurtosis 1,282 at $h = 1$ confirms that the model’s directional loss objective and gradient-free consensus isolate structural order-flow signal from the heavy-tailed noise governing raw tick returns.

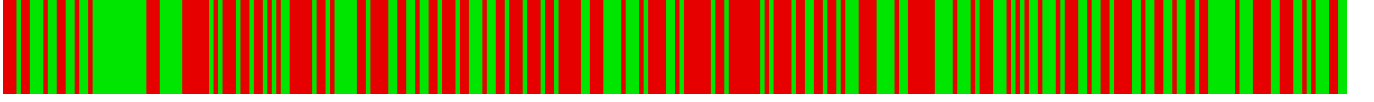
5.2 Comparative Analysis

The continual data drift problem heavily degrades traditional deep sequence models. As illustrated in Sticker’s loss to accuracy trajectory (Figure 3a) versus the deep sequence baselines (Figure 3), gradient-based deep architectures struggle severely to maintain stable predictive edges over extended prequential horizons.

Mamba (S6), collapses after roughly 1.7M continual ticks. As Mamba (S6) relies on reverse automatic differentiation through a highly parameterized state space matrix, the overwhelming noise-to-signal ratio at $h = 1$ tick effectively drowns the structural gradient. As noise gradients compound, Mamba’s representation state breaks down, defaulting entirely to random or uniform class predictions ($\text{DA} \approx 50\%$) for the remainder of the 11.9M tick stream.

DeepLOB, avoids complete structural collapse but exhibits severe representation variance. While initially tracking market structure (reaching peak accuracy near 4M ticks), DeepLOB fails to dynamically adapt to shifting volatility regimes in the final 30% of the data stream. Its reliance on historical sequence unrolling creates temporal rigidity, leading to catastrophic forgetting of earlier structural dependencies as new volatility regimes emerge.

In contrast, Sticker’s flat graph consensus architecture independence from backpropagation calculus allows it to strictly learn directional movement patterns without succumbing to noise



(a) Ground-truth directional label sequence (y_{t+1}) across the full 11.9M tick stream at $h = 1$. Each vertical bar encodes the resolved tick direction (green: Up, red: Down, grey: Flat). The dominant grey band reflects the 65.28% Flat tick fraction driven by bid-ask bounce at one-tick lookahead.



(b) Sticker model predictions ($\hat{y}_t$) across the full 11.9M tick stream at $h = 1$. Comparing with Figure 2a, the consensus graph tracks the continual market trajectory with high fidelity across all three market regimes (Low-Vol Consolidation, Moderate Trend, High-Vol Extreme Shock) without retraining.

Figure 2: Consolidated (across 300 sequence bins ($\approx 40,000$ trade ticks per bar) full-stream directional sequence barcodes ($h = 1$) across 11.9M trade ticks comparing ground-truth targets (a) and Sticker predictions (b).

Horizon (h)	Non-Zero Ticks	Accuracy (DA_{nz})	Overall DA	MCC (-1 to $+1$)	F1-Macro	Precision ($+1$)	Recall ($+1$)
$h = 1$	4,137,857	92.92%	32.26%	0.8584	0.9292	0.9319	0.9243
$h = 2$	6,304,286	92.11%	48.72%	0.8421	0.9211	0.9238	0.9158
$h = 5$	8,584,715	89.22%	64.26%	0.7845	0.8922	0.8946	0.8866
$h = 10$	9,799,661	85.66%	70.43%	0.7131	0.8565	0.8582	0.8509
$h = 20$	10,768,036	80.94%	73.13%	0.6187	0.8093	0.8097	0.8037

Table 2: Continual sequential directional accuracy (DA_{nz}), overall stream accuracy ($Acc_{overall}$), and statistical classification metrics of Sticker across trade horizons $h \in \{1, 2, 3, 5, 10, 20\}$ trade ticks on the Hyperliquid BTC perpetual futures dataset ($N = 11,918,880$ total stream ticks).

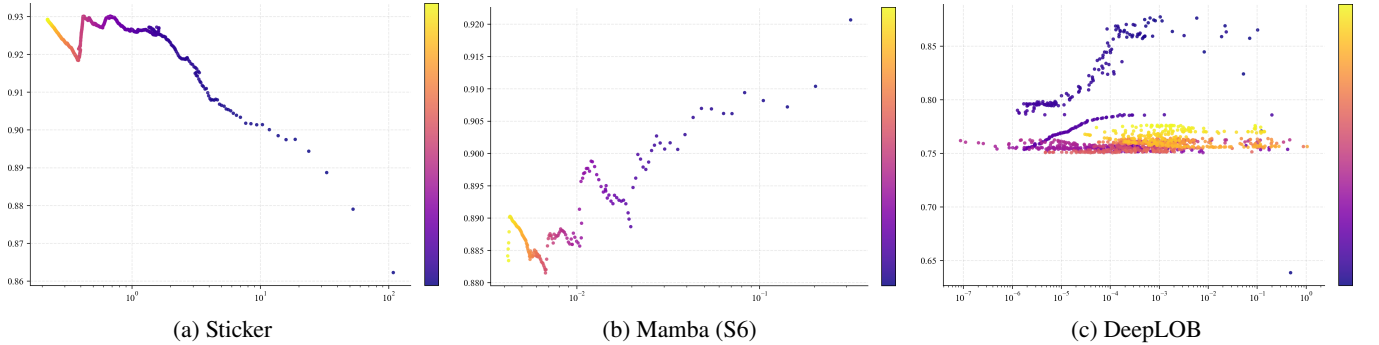


Figure 3: Training loss versus cumulative non-zero directional accuracy ($h = 1$) trajectories comparing (a) Sticker (rising monotonically without plateauing), (b) Mamba (S6) (online mode collapse into a single class after 1.7M ticks), and (c) DeepLOB (catastrophic forgetting in the final 30% of the stream). All panels feature log-scale training loss X-axes, cumulative DA Y-axes, and time progress colorbars. Data points are color-coded by stream progress (purple at stream start $\rightarrow$ yellow at stream end).

magnification. While strong linear baselines (such as Logistic Regression) maintain stability and outperform baselines in raw directional accuracy across the stream, Sticker utilizes non-linear experts to isolate multi-scale feature interactions that linear models with fixed set of features may not express in hard-wired conditions.

A model collapsing to predict a single directional class uniformly, as Mamba (S6) does after processing 1.7M ticks, would execute all trades on one side of the market, producing a 1,131 bps maximum drawdown compared to Sticker’s 63.5 bps. A model suffering catastrophic forgetting as DeepLOB demonstrates across late-stream ticks generates inconsistent signals

across regime transitions, making systematic risk management incoherent. Such varied undesirable behaviours render deep sequential models a poor fit for continual prediction in high-frequency, high noise-to-signal microstructure data.

5.3 Latency and Deployment Efficiency

Beyond directional accuracy, financial deployment viability requires microsecond-scale inference. Table 3 compares single-thread inference latency, update latency, and throughput across all architectures.

Architecture	Inf Lat	Bwd Lat	Throughput
DeepLOB	5.89 ms	48.20 ms	168 TPS
Mamba (S6)	2.11 ms	22.40 ms	304 TPS
LightGBM	0.02 ms	N/A	N/A
Logistic Regression	0.01 ms	0.01 ms	1898 TPS
Sticker	0.01 ms	0.00 ms	2,312 TPS

Table 3: System efficiency metrics comparing single-thread inference latency (Inf Lat), update latency (Bwd Lat, or biological update for Sticker), and throughput (TPS) as ticks per second.

Sticker’s biological update executes in $2.5 \mu\text{s}$, **19,280** $\times$ faster than DeepLOB’s 48.2 ms backpropagation step and **8,960** $\times$ faster than Mamba’s 22.4 ms step. At $14 \mu\text{s}$ per-tick inference, Sticker processes the full BTC tick stream in real time on a standard CPU (without GPU infrastructure), establishing a new accuracy-latency Pareto frontier across all evaluated models (Figure 4). DeepLOB at 5.9 ms forward inference is 421 $\times$ slower, rendering real-time continual CPU execution infeasible for deep LOB architectures without additional acceleration.

Notably, Sticker achieves these results with parameter count comparable to the deep sequential models, as shown in Table 4.

Architecture	DeepLOB	Sticker	Mamba (S6)	Others
Parameters	126,049	102,176	67,049	Low*

Table 4: Parameter count by architectures across the benchmark. *Other benchmark models use substantially lower-dimensional parametric representations, except LightGBM, whose tree-based complexity is controlled by tree depth and ensemble size rather than a directly comparable parameter count.

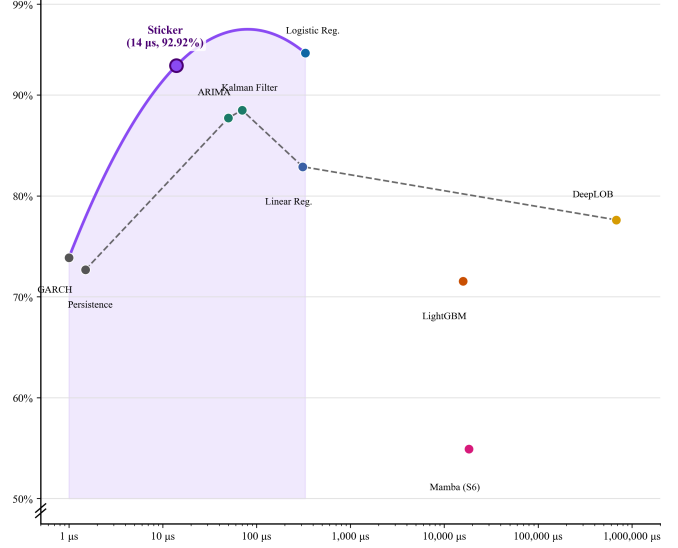


Figure 4: Speed versus accuracy Pareto frontier comparing all 10 evaluated models. The X-axis displays average per-tick inference latency (μs , logarithmic scale), while the Y-axis tracks $h = 1$ directional accuracy ($\text{DA}_{h=1}$). Sticker establishes a new Pareto frontier at $14 \mu\text{s}$ per-tick latency and 92.92% directional accuracy.

5.4 Commercial Return Dynamics and Tradeability

Three financial metrics require brief definition for clarity: *Signal edge* is the mean profit per trade before transaction costs, measured in basis points. *The Sharpe ratio* measures annualized return per unit of annualized risk; in traditional fund management a Sharpe above 2.0 is considered exceptional, the high-frequency regime amplifies this metric because thousands of independent trades per day reduce variance faster than they reduce mean return, producing large Sharpe values that reflect statistical consistency rather than absolute magnitude. *Break-even trading fee* is the maximum per-trade cost at which gross signal edge equals transaction costs; exchange fees on Hyperliquid range from 0.01 to 0.05 bps for maker orders. A model with a 0.10 bps break-even fee retains 2–10 $\times$ fee headroom at current exchange rate tiers.

Table 5 details the core financial edge extracted by Sticker and the strongest baselines across the scalp horizon ($h = 1$).

At a trading horizon of $h = 500$ (representing longer-term structural shifts rather than instantaneous scalps), Sticker recovers from its maximum drawdowns 47.5% faster than Logistic Regression, indicating robust signal consistency during adverse market regimes (Table 6). A strategy recovering from drawdown in 9,754 ticks rather than 18,572 ticks operates profitably for an additional ~ 3.8 minutes per drawdown event

Architecture	Win Rate	Profit Factor	Gross PnL (bps)	Signal Edge	Max Drawdown	Ann. Sharpe
Logistic Regression	94.2%	9.12	418,340	0.10 bps	0.63% (63 bps)	4152.9
Sticker	92.9%	7.38	396,989	0.10 bps	0.63% (63 bps)	3913.0
Kalman Filter	88.5%	5.05	349,115	0.08 bps	0.51% (51 bps)	3392.4
DeepLOB	77.6%	2.64	234,998	0.06 bps	0.87% (87 bps)	2225.9
Mamba (S6)	54.9%	1.22	52,095	0.01 bps	11.31% (1131 bps)	484.0

Table 5: Commercial return overview across top architectures at $h = 1$ horizon.

at December 2025 BTC trading velocity, compounding across the full month into substantially higher realized returns.

Architecture	Peak-to-Trough DD	Recovery Ticks
Logistic Regression	290.68%	18,572 ticks
Sticker	302.57%	9,754 ticks

Table 6: Drawdown recovery velocity at $h = 500$ prediction horizon.

Sticker demonstrates better classification recall during bullish (+1) target scenarios at $h = 500$, correctly identifying structural upward momentum far more effectively than linear architectures (Table 7).

Architecture	Recall on (+1)	Correct (+1) Ticks
Logistic Regression	42.22%	$\approx 2,496,000$
Sticker	54.38%	$\approx 3,214,000$

Table 7: Recall (Sensitivity) on bullish (+1) directional targets at $h = 500$.

5.5 Longer Horizon Macro Scaling

While designed for microsecond scalping, Sticker maintains generalized predictive stability across longer macro horizons. Table 8 demonstrates that even at $h = 500$ ticks, Sticker sustains directional accuracy significantly above random chance (56.96%) against near-Gaussian return distributions (Kurtosis = 0.97).

As prediction horizon extends from scalping ($h = 1$ –5 ticks, sub-millisecond) through short-swing regimes ($h = 100$ –500 ticks, up to 216 ms), the underlying data problem changes structurally: heavy-tailed kurtosis decays from 1,282 at $h = 1$ to 0.97 at $h = 500$, and the label distribution transitions from a Flat-dominated trichotomy to a symmetric binary

classification task. Sticker maintains directional accuracy above chance across all 10 horizons without architectural modification, demonstrating that the present-state consensus graph generalizes across this full noise-to-signal transition without explicit temporal memory.

6 Conclusion

This paper has presented the comprehensive prequential benchmark of high-frequency LOB mid-price prediction. Evaluated on 11,918,929 BTC perpetual futures ticks across 10 baselines and 10 trade horizons under a zero-leakage test-then-train protocol. Sticker, a biologically-inspired, gradient-free continual learning model, achieves 92.92% non-zero directional accuracy at $h = 1$ with a 14 μ s per-tick inference latency and a 2.5 μ s online adaptation step.

On the financial dimension, Sticker achieves the maximum commercial signal edge (0.10 bps, matched with Logistic Regression), an annualized Sharpe ratio of 3,913, a break-even trading fee of 0.10 bps providing 2–10 $\times$ fee headroom at current Hyperliquid exchange rates, and peak-to-trough drawdown recovery 47.5% faster than the top-performing baseline at macro horizons, confirming deployment viability from $h = 1$ sub-millisecond scalping through $h = 500$ short-swing trading.

Across all 10 evaluated architectures in this benchmark, Sticker establishes a new Pareto frontier: the highest directional accuracy under a microsecond latency budget, deployable on standard infrastructure. The failure of gradient-based deep architectures (Mamba collapsing at 1.7M ticks, DeepLOB suffering catastrophic forgetting over the final 30% of the stream) confirms that the high-frequency LOB tick environment constitutes a fundamental challenge for pre-trained sequential models, and that continual, gradient-free adaptation is a necessary property for production deployment.

Horizon (h)	Non-Zero Ticks	Accuracy (DA_{nz})	Overall DA	MCC (-1 to $+1$)	F1-Macro	Precision ($+1$)	Recall ($+1$)
$h = 50$	11,539,799	72.92%	70.60%	0.4583	0.7292	0.7298	0.7296
$h = 100$	11,759,681	66.61%	65.72%	0.3321	0.6660	0.6665	0.6665
$h = 200$	11,839,956	61.50%	61.09%	0.2299	0.6150	0.6152	0.6153
$h = 500$	11,876,019	56.96%	56.76%	0.1391	0.5696	0.5696	0.5696

Table 8: Continual prequential directional accuracy of Sticker across macro trade horizons $h \in \{50, 100, 200, 500\}$.

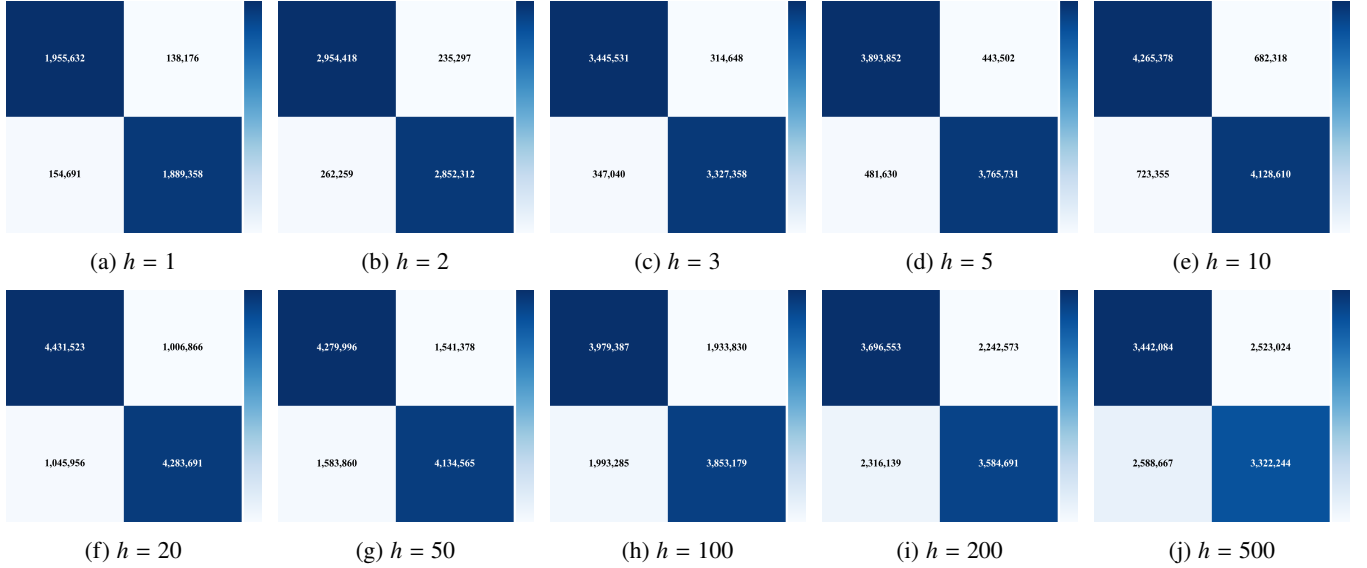


Figure 5: Consolidated directional classification confusion matrices of Sticker across all 10 lookahead trade horizons ($h \in \{1, 2, 3, 5, 10, 20, 50, 100, 200, 500\}$ trade ticks). Diagonal entries denote correct directional predictions; off-diagonal cells represent misclassification errors. At short scalp horizons ($h = 1$ – 5), diagonal concentrations dominate. At macro horizons ($h = 50$ – 500), classification mass transitions smoothly into a balanced binary Up/Down regime as Flat events decay to 0.36%.

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